3. Provide the requested information for the **Folders** tab as follows:
   - **Export Folder** box—Type the path to the folder in which all of the exported test data will reside. Alternatively, the default path supplied can be used.
   - **Report Folder** box—Type the path to the folder in which all of the reports will reside. Alternatively, the default path supplied can be used.
   - **Backup Folder** box—Type the path to the folder in which the backup database will reside. Alternatively, the default path supplied can be used.
   - **DB Log Folder** box—Type the path to the folder in which the database log files will reside. Alternatively, the default path supplied can be used.

Caution

The default locations for each of the folders is located on the computer hard drive. To guard against loss of data, the files in the export folder should be periodically copied to a different computer or server. If the GeneXpert Dx system is connected to a network, it is possible to archive files directly to a server.

4. Click **OK** to save the changes and close the window.

2.11.4 Xpress Settings Tab

The **Xpress Settings** tab is not to be used in the normal operation of the system. Do not make any changes to the settings on this tab.

2.11.5 Host Communication Settings Tab

The **Host Communication Settings** tab is used to configure the system software when a GeneXpert Dx is connected to a Laboratory Information System (LIS) host computer or Cepheid Link.

Note No configuration of this tab is required if an LIS system is not being used with the system.

Note To configure the host communication settings for an LIS, see Section 2.11.5.1, Configuring Host Communications for an LIS. To configure the host communication settings for Cepheid Link, see Section 2.11.5.2, Configuring Host Communications for Cepheid Link.

2.11.5.1 Configuring Host Communications for an LIS

1. In the GeneXpert Dx System window (see Figure 2-29), click **Setup** on the menu bar, then click **System Configuration** (see Figure 2-30).
2. Click the **Host Communication Settings** tab. The **Host Communication Settings** tab appears. See Figure 2-55.

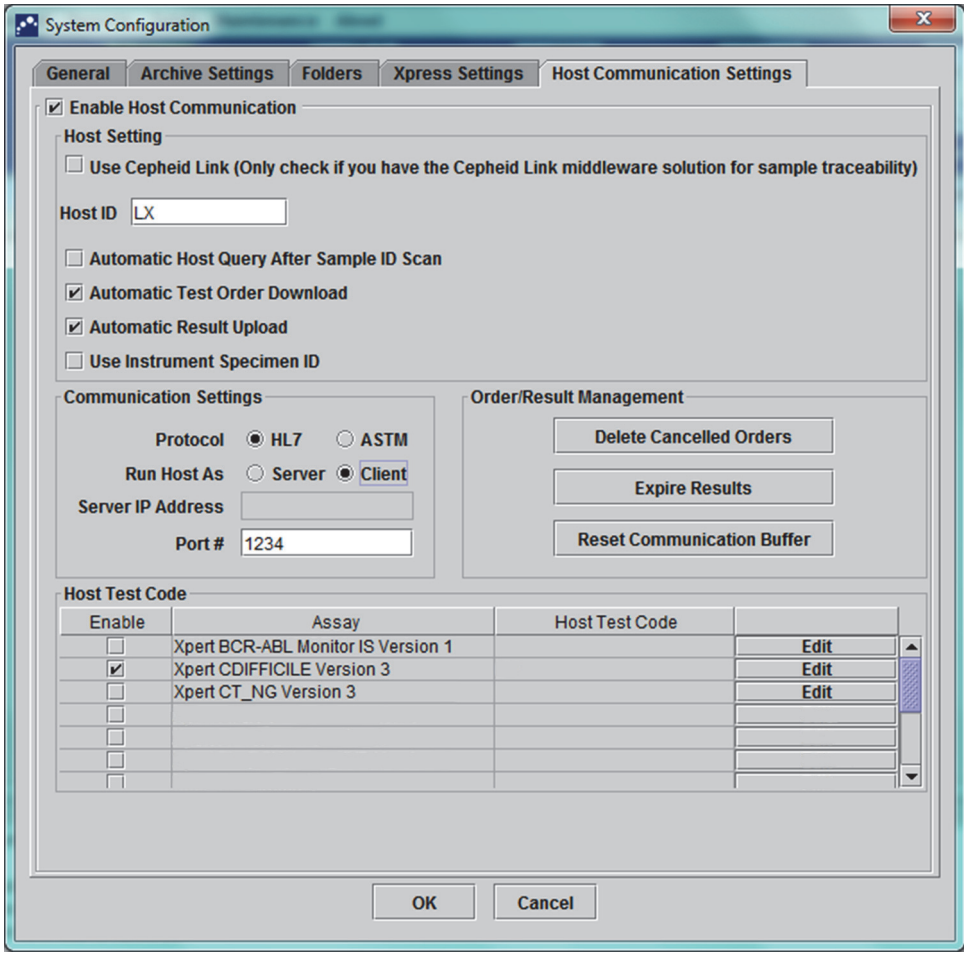

Figure 2-55. System Configuration Dialog Box (Host Communication Settings Tab)

Note

If the LIS is being enabled on a new system, there will be no assays shown.

Caution

Within the hospital or laboratory network, each GeneXpert Dx system should have a unique system name, which is used for host communication. The LIS host administrator should control the process for defining system names.

Important

Do not check the Use Cepheid Link checkbox when configuring the host communication settings for a hospital LIS system.

3. Provide the settings to configure the communication between the GeneXpert Dx software and a Laboratory Information System (LIS):
   - **Enable Host Communication**—Select to enable the GeneXpert Dx software connected to a host. Clear to disable the host communication.
     - **Host ID**— Type in a unique host name to identify an LIS or Data Management System (DMS) that is connected to this GeneXpert Dx system. The maximum number of characters is 20.
     - **Automatic Host Query After Sample ID Scan**—Select to enable the GeneXpert Dx system to query for test orders associated with the scanned or entered Sample ID.
     - **Automatic Test Order Download**—Select to enable the GeneXpert Dx system to periodically query all test orders from the host.

Caution

If the Host is connected to multiple GeneXpert Systems, you may want to:
• Use Automatic Host Query After Sample ID Scan instead of Automatic Test Order Download to minimize duplicate orders to multiple GeneXpert systems.
• The host should download order to a specific GeneXpert System.
• If orders are sent to multiple GeneXpert systems, the host should cancel pending orders when completed result is received.

     - **Automatic Result Upload**—As soon as the test is completed, the results are uploaded.
     - **Use Instrument Specimen ID**—Select to enable the GeneXpert Dx system to generate a unique specimen ID, which is returned to the host. The Instrument Specimen ID is a unique ID for this sample. It should be stored in the host and used for future communication for this sample. This option is applicable if the facility does not provide unique sample identification.

       If the facility provides unique sample identification, this setting should be disabled.
     - **Communication Settings** box—Select or clear the following check boxes:
       - **Protocol**—Select HL7–compatible or ASTM–compatible protocol.
       - **Run Host As**—For socket connection between the two systems. Select to run the host as a server or a client.
       - **Server IP Address**—If **Run Host As Server** option is selected, an IP address with 4-part value (N.N.N.N) should be entered. The value should match the IP address of the host server. N is between 0-255. If **Run Host As Client** option is selected, the IP address of the network card available for host connectivity is displayed.
       - **Port #**—The port number should be between 1024 to 65535.

Caution

The network port that is dedicated for the GeneXpert instrument should not be used for the host connection. The second NIC available on each GeneXpert computer should be used to connect the GeneXpert Dx system to the host.

- **Order/Result Management**—Click the appropriate buttons:
  - **Delete Canceled Orders**—Click to delete canceled orders. This is useful to remove redundant orders during host communication testing.
  - **Expire Results**—Click to expire results pending upload for tests that should no longer upload to the host.

Caution

Do not use Reset Communication Buffer (discussed below) during normal operation; otherwise, you will have to re-download orders and re-upload results.

- **Reset Communication Buffer**—To clear the data between the GeneXpert Dx system and the host. This is useful to remove data during host communication testing.
- **Host Test Code** table—This look up table allows the host administrator to type in the test code that was entered into the host, so it can be translated into the GeneXpert Dx system for test order processing and result reporting.
  - **Enable**—Indicates if the assay has been set up for test order download and result reporting.
  - **Assay**—Assay name available for host connectivity.
  - **Host Test Code**—the test code which the host used for download of test order and upload of test result.

Important

You cannot edit the test code for old versions of an assay. If you update the test code, the update will only apply to the new version of the assay; therefore, you must change the test code before upgrading an assay.

Caution

Be careful to not use the same test code for tests from two different assays.

4. Click **Edit** button to enable the assay for host use and to define host test codes for that assay. See Section 2.11.6 to configure the assay for order and result upload and to define host test codes.
5. Click **OK** to save the changes and close the window.

2.11.5.2 Configuring Host Communications for Cepheid Link

Important

Once the system has been configured for Cepheid Link, it cannot be used for non-LIS originated test orders or for running external controls without disabling Cepheid Link. Cepheid Link may be enabled again after running non-LIS originated test orders or external controls.

To enable and configure host communications for the GeneXpert Dx systems to Cepheid Link:

1. On the GeneXpert Dx system window (see Figure 2-29), select the **SETUP** button, then select the **SYSTEM CONFIGURATION** button (see Figure 2-30**).**
2. Select the **HOST COMMUNICATIONS SETTINGS** button (see Figure 2-30**)** to display the Host Communications Settings workspace. See Figure 2-55.
3. To enable host communication, select the **Enable Host Communication** check box in the upper left corner of the workspace (see Figure 2-56). This allows other options to be selected on the Host Communication Settings screen.

Important

Within the hospital or laboratory network, each GeneXpert system should have a unique system name which is used in the communication. The host administrator should control the process for defining system names.

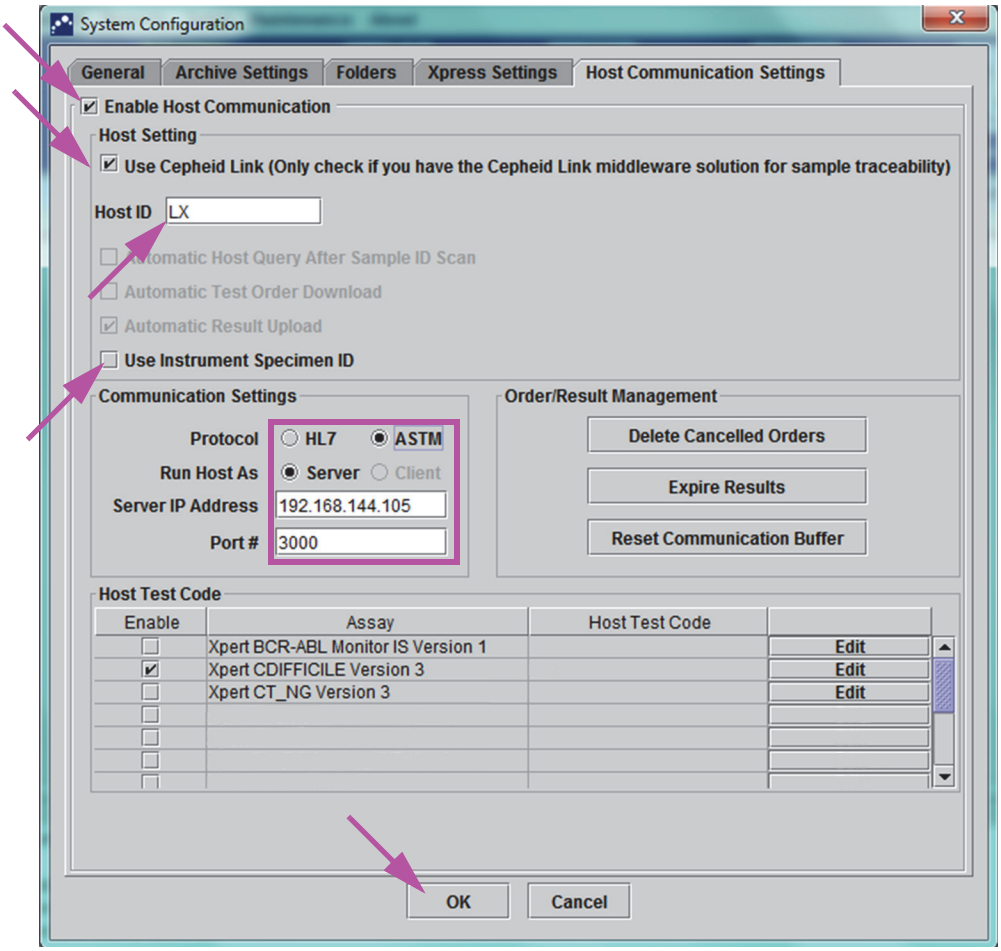

Figure 2-56. Host Communication Settings Workspace Configured for Cepheid Link

Important **All of the information to be entered into this workspace must be provided by the LIS network administrator. It is not provided by Cepheid.**

4. Select the **Use Cepheid Link** checkbox to set up the host communication for Cepheid Link. After selecting the **Use Cepheid Link** checkbox, most of the configuration will be automatically set up. See Figure 2-56.
5. In the General section of the Host Communication Settings workspace, enter the appropriate information and select the appropriate items to interface with the LIS network.
   - **Host ID** field – Type in a unique host name to identify this GeneXpert Dx system. The maximum number of characters is 20.
   - **Automatic Host Query After Sample ID Scan** check box – This check box is disabled when connecting to Cepheid Link,
   - **Automatic Test Order Download** check box – This check box is disabled when connecting to Cepheid Link,
   - **Automatic Result Upload** check box – This check box is enabled when connecting to Cepheid Link,
   - **Use Instrument Specimen ID** check box – Select to enable the GeneXpert system to generate a unique specimen ID which is returned to the host. The Instrument Specimen ID is a unique ID for this sample. It should be stored in the host and used for future communication for this sample. This option is applicable if the facility does not provide unique sample identification.

     If the facility provides unique sample identification, this setting should be disabled.
6. In the Protocol section of the Host Communication Settings workspace, select either **HL7**-compatible or **ASTM**-compatible protocol.
7. In the Communication Settings section of the Host Communication Settings workspace, the host must be set to **Server** to communicate with Cepheid Link.
   - **Server IP Address** field – An IP address with 4-part value (**N.N.N.N**) should be entered. The value should match the IP address of the Cepheid Link server. **N** is between 0-255.
   - **Port #** field – The port number must be **3000** to communicate with the Cepheid Link server.
8. After you have set up the host communications for the Cepheid Link server, select the **OK** button. See Figure 2-56.

   Select the **Cancel** button if you do not want to save the host communication settings.

Note

Cepheid recommends to always confirm that LIS or HIS uploaded results match GeneXpert results after any changes to the GeneXpert or host system, including, but not limited to, changes to the following:
- GeneXpert software version
- GeneXpert assay definition files and version
- GeneXpert host communication settings
- Host middleware software or configuration changes
- LIS software or configuration changes

2.11.6 Configuring Assay for Order and Result Upload

Caution

In order to perform the required assay, the same test code should be entered in the host, the GeneXpert Dx system and the Cepheid Link system, if applicable.

Caution

Do not change test orders until all test results have been uploaded.

2.11.6.1 Configuring a Single-Result Assay for Order and Result Upload

1. **In the Host Test Code** table section of the Host Communication Settings tab (see Figure 5-88), click the desired **Edit** button to change the setting. The Define Test Code dialog box appears. See Figure 2-57.

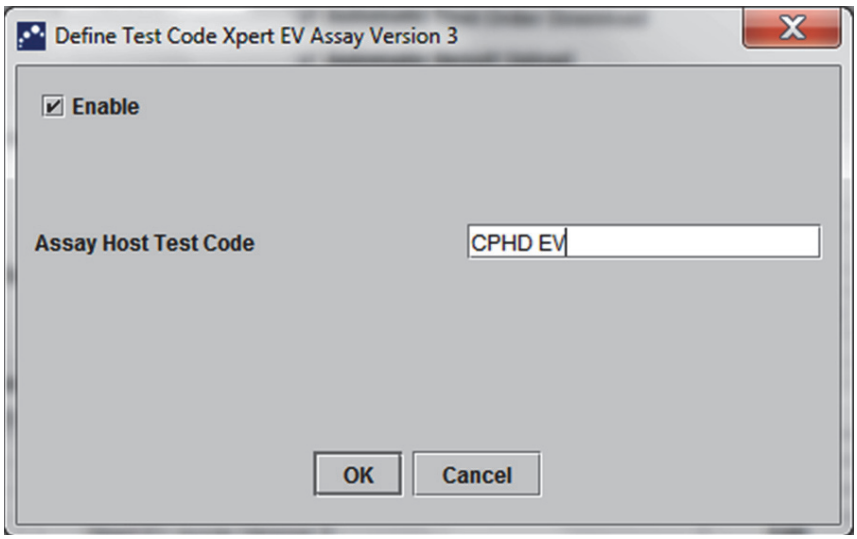

Figure 2-57. Define Test Code Dialog Box for a Single-Result Assay

2. Click the **Enable** checkbox to allow the host to download test orders and GeneXpert Dx system to upload results to the host using the defined assay test code.
3. In the **Assay Host Test Code** field of the Define Test Code dialog box, enter the same test code that was entered into the host system and the Cepheid Link system, if applicable.(the test code entered for the GeneXpert Dx system must be the same as